Kirandeep Kaur

Paul G. Allen School of Computer Science & Engineering
University of Washington, Seattle

kaur13@cs.washington.edu
Website
LinkedIn @ikiran013

Education

2023 – present **University of Washington.** **Seattle, WA**
Ph.D. in Computer Science & Engineering

2020 – 2022 **Indian Institute of Technology (IIT).** **Ropar, India**
M.Tech. in Computer Science
Advisors: *Shweta Jain*, *Sujit Gujar*

Grants and Awards

2026 Community-Centered Computing Initiative's (CECI) Grant
Runner Up- **People's Choice Award** Allen School (Undergrad) Research Showcase.
AWS Cloud Computing **Research Credit Award**.
ACM CHIIR **Scholar Grant**.
2025 Nominated for the **Excellence in Teaching Award**.
2024 ACM RecSys **Scholar Grant**.
Microsoft Endowed Fellowship, Microsoft.
2022 The Cornell, Maryland, Max Planck **Pre-Doc Fellowship**.
Nominated for **Best MTech Thesis Award** at IIT Ropar.
ACM KDD **Scholar Grant**, IIT Gandinagar.
2020-2022 **Indira Gandhi Scholarship**, IIT Ropar.
2015-2019 **Post-Matric Scholarship for Undergraduate Studies**, Guru Nanak Dev University.

Research Publications

* and † denote equal contribution; **Google Scholar**

Ongoing Work

[M.1] From Epistemic Authority to Moral Status: How People Form Models of Possible Digital Minds?

[M.2] Value Sensitive Design for Proactive AI Doctor.

[M.3] Analyzing Proactive Information Seeking Behavior and Attack Surfaces in Complex Agentic Frameworks.
People's Choice Award *runner up, Allen School Undergraduate Research Showcase*

Journal Publications

[J.1] **K.Kaur**, Vinayak Gupta, Manya Chadha, and Chirag Shah.
Efficient & Responsible Adaptation of Large Language Models for Top-k Recommendations.
> *ACM Transactions on Recommender Systems*, 4(3), Article 41, 2026.
> *Generative Models for Recommender Systems Workshop (ROEGEN@RECSYS'24).*

[J.2] Shivam Gupta, **K.Kaur**, and Shweta Jain.
EqBal-RS: Mitigating Popularity Bias in Recommender Systems.
> *Journal of Intelligent Information Systems*, 62(2), 509–534, 2024.

Conference Proceedings

[C.1] **K. Kaur**, Xingda Lyu, and Chirag Shah.
Knowing Isn't Understanding: Re-grounding Generative Proactivity with Epistemic and Behavioral Insight.
Forty-third International Conference on Machine Learning, ICML 2026.
Human-Centered Proactive Agents for Interactive Information Access Workshop @CHIIR'2026.
AI-CoScientist Workshop @ARD, Amazon Research Day 2026.
Featured under "Epistemic Incompleteness and the Proactivity Paradox" by Prompted LLC.
Featured in Machine Brief News.
Featured under "Proactive Knowledge Behaviors Framework" by Emerging Minds

[C.2] **K. Kaur***, Vinayak Gupta*, Aditya Gupta, and Chirag Shah.
PROPER Agents: Proactivity Driven Personalized Agents for Advancing Knowledge Gap Navigation.
> *The 64th Annual Meeting of the Association for Computational Linguistics.* ACL 2026.
> *Human-Centered Proactive Agents for Interactive Information Access Workshop* @CHIIR'2026.
> *AI-CoScientist workshop @ARD,* Amazon Research Day 2026.
Featured under "ProPerAssistant: Proactive & Personalized AI" by Emerging Minds

[C.3] **K.Kaur**, Preetam Daammu, Hideo Joho, and Chirag Shah.
Beyond Static Evaluation: Rethinking the Assessment of Personalized Agent Adaptability in Information Retrieval.
> In *Proceedings of the 2025 Annual International ACM SIGIR Conference on Research and Development in Information Retrieval in the Asia Pacific Region*, SIGIR-AP 2025.

Workshop Papers & Technical Reports

[W.1] **K. Kaur**, and Amit Goyal.
Cold-Start Audiobook Recommendation via Cross-Domain Sub-Tower Fusion.
> *Generative AI for Streaming Media GenAI4SM Workshop*@WSDM 2026.
> *Graph based Machine Learning GraphML Workshop*@AMLC 2025.

[T.1] **K. Kaur**, Sujit Gujar, and Shweta Jain.
Towards Fairness in Provably Communication-Efficient Federated Recommender Systems.
> *Technical report ,* IIT Ropar, 2022.

[T.2] **K. Kaur**, Alessandro Fabris, and Asia Biega.
Investigating the Importance of Data Minimization in the Era of Federated Learning.
> *Technical report ,* Max Planck Institute for Security and Privacy, 2023.

Professional Experiences

Jun 2026 – Sept 2026 **Amazon Science**. **Bellevue, WA**
Applied Scientist II Intern
Science Intern at Applied AI team with Jing Yang, Hitesh Arora, Chirag Shah, *Francesco Carbone.*
Project: Collective Experiential Memory for Proactive Multi-Agent Systems.

Jun 2025 – Sept 2025 **Amazon Science**. **San Francisco, CA**
Applied Scientist II Intern
Science Intern at Amazon Music with *Amit Goyal.*
Project:Cold-start audiobook recommendation via cross-domain transfer learning.

Jan 2023 – Sept 2023 **Max Planck Institute for Security & Privacy**. **Bochum, Germany**
Visiting Researcher.
Research visitor at the Responsible AI Group with *Asia Biega.*
Investigated how GDPR's data minimization principle maps to Federated Learning frameworks.

Aug 2021 – Sept 2021 **Alan Turing Institute & Rosalind Franklin Inst.**. **London, UK**
Participating Researcher.
Participated in Data Study Group and collaborated with a team of 9.
Improved the resolution of protein structure imaging via dark-frame subtraction to reduce noise.

Teaching Experience

Instructor

Fall 2025 **Course Instructor**, INFO 376: Intro to Search and Recommendation Systems, UW
Re-designed and taught an undergraduate class (of 35) on search and recommendation systems. Incorporated modern and emerging Generative IR techniques, designed assignments, exams, and in-class activities. Held weekly office hours and managed a teaching assistant. Mentored ten class projects and provided feedback throughout the quarter.

Teaching Assistant

Spring 2026 **Teaching Assistant**, CSE 493G1, Deep Learning for Computer Vision, UW
Instructor: Ranjay Krishna

Winter 2026 **Teaching Assistant**, CSE 473, Artificial Intelligence, UW
Instructor: Taylor Kessler Faulkner

Spring 2025 **Teaching Assistant**, CSE 493G1, Special Topics: Introduction to Deep Learning, UW
Instructor: Ranjay Krishna

Guest Lectures and Panels

Spring 2026 **Presenter**, Computational Minds and Machines lab
Invited by: Max Kleiman-Weiner
Talk: Regrounding Proactive Assistance for Effective Human-AI Collaboration

Spring 2026 **Panelist**, CSE494: Guided Research in Computer Science and Engineering, UW
Served as a Panelist alongside multiple PhD students to answer questions of undergraduates on mentorship and research in computer science.

Spring 2025 **Guest Lecturer**, CSE 581: Computer Ethics, UW
Delivered a lecture on privacy and moral ethics in AI systems.

Summer 2025 **Panelist**, Porsesh Policy Research Institute, WA
Served as a Panelist alongside Dr. Bill Howe at the AI for Seniors event hosted by Ehsan Shayegan, Porsesh Policy Research Institute.

Mentorship Experience

Masters Students

2026 – *present* **Ziyu Zhou**, UW ISchool
Epistemic Intelligent Machines for Proactive Assistance.

2026 – present **Priyanshi Garg**, UW Linguistics.
Value Sensitive Design for Proactive Agents.

2024 – 2025 **Manya Chadha**, UW ISchool → Data Scientist at Kaizen Analytics.
Responsible Adaptation of LLMs for Recommendations.

Undergraduate Students

2025 – *present* **Xingda Lyu**, BS Student (Stat & Informatics)
Social-cognitive learning for grounding epistemic reasoning in LLMs.

2026 – *present* **Kevin Nguyen**, BS Student at (Computer Science)
Information flow and attack surfaces in complex agentic frameworks.
People's Choice Award *runner up, Allen School Undergraduate Research Showcase*

2026 – *present* **Akshith Sravanan**, BS Student (Computer Science)
Information flow and attack surfaces in complex agentic frameworks.
People's Choice Award *runner up, Allen School Undergraduate Research Showcase*

2026 – *present* **Jasmine Marwaha**, BS Student (Information Science)
Value Sensitive Design for Proactive Agents.

High School Students

2024 – 2025 **Aditya Gupta**, Issaquah High School → Undergrad at Yale University
Proactive Agents for Knowledge Gap Navigation.

Professional Service

Organizing Committees

2026 **Women In Machine Learning (WIML) workshop**, COLM 2026
2026 **Human-Centered Proactive and Personalized Agents Workshop**, ACM CHIIR 2026
2024 **IEEE Innovation Renaissance Event**, IEEE Seattle Section

Student Volunteer Chair

2026 **ACM SIGIR Conference on Human Information Interaction and Retrieval**, ACM CHIIR 2026

Reviewing

Conf. & Workshops	**NeurIPS** 2026, **NeurIPS Ethics Reviewer** 2026, **SocialSim @ COLM** 2026, **WAB @ COLM** 2026, **GenAI4SM @ WSDM** 2026, **IASEAI** 2026
Journal	**TOIS** (ACM Transactions on Information Systems) 2025, **IJAI** (International Journal of Artificial Intelligence) 2026

Community Service

2025 – present	**Co-founder**, Networking Series at RAISE, University of Washington *Co-founded quarterly in-person networking and unconference session series (RAISE Fall Exposition 2025, RAISE AI for ALL Event 2025.)*
2025 – present	**Co-Organizer**, RAISE Responsible AI Webinar Series, University of Washington *Coordinating weekly research talks and discussions on responsible AI.*
2025 – present	**Mentor**, UW Allen School Pre-Application Mentorship Service (PAMS). *A program supporting potential CS PhD applicants, with 80% from underrepresented communities.*
2024	**Mentor**, UW Allen School First-Year PhD Mentorship Program. *Co-mentored two first-year PhD students by conducting weekly in-person mentorship sessions and monthly outings.*
2024	**Student Volunteer**, ACM RecSys 2024. *Supported conference logistics and community engagement.*
2021	**Student Volunteer**, IWDG,2021. *Supported virtual sessions at International Workshop on Domination in Graph*
2020-2022	**Teaching Volunteer**, IITRpr. *Taught Maths and English to underprivileged students at an NGO - Pehchaan Ek Safar.*